Industrial, not fruit fructose intake is associated with the severity of liver fibrosis in genotype 1 chronic hepatitis C patients.

BACKGROUND & AIMS: Unhealthy food intake, specifically fructose, has been associated with metabolic alterations and with the severity of liver fibrosis in patients with non-alcoholic fatty liver disease. In a cohort of patients with genotype 1 chronic hepatitis C (G1 CHC), we tested the association of fructose intake with the severity of liver histology. METHODS: Anthropometric and metabolic factors, including waist circumference (WC), waist-to-hip ratio (WHR), dorso-cervical lipohypertrophy and HOMA were assessed in 147 consecutive biopsy-proven G1 CHC patients. Food intake, namely industrial and fruit fructose, was investigated by a three-day structured interview and a computed database. All biopsies were scored by an experienced pathologist for staging and grading (Scheuer classification), and graded for steatosis, which was considered moderate-severe if ≥ 20%. Features of non-alcoholic steatohepatitis (NASH) in CHC were also assessed (Bedossa classification). RESULTS: Mean daily intake of total, industrial and fruit fructose was 18.0±8.7g, 6.0±4.7g, and 11.9±7.2g, respectively. Intake of industrial, not fruit fructose, was independently associated with higher WHR ($p=0.02$) and hypercaloric diet ($p<0.001$). CHC patients with severe liver fibrosis (■F3) reported a significantly higher intake of total (20.8±10.2 vs. 17.2±8.1g/day; $p=0.04$) and industrial fructose (7.8±6.0 vs. 5.5±4.2; $p=0.01$), not fruit fructose (12.9±8.0 vs. 11.6±7.0; $p=0.34$). Multivariate logistic regression analysis showed that older age (OR 1.048, 95% CI 1.004-1.094, $p=0.03$), severe necroinflammatory activity (OR 3.325, 95% CI 1.347-8.209, $p=0.009$), moderate-severe steatosis (OR 2.421, 95% CI 1.017-6.415, $p=0.04$), and industrial fructose intake (OR 1.147, 95% CI 1.047-1.257, $p=0.003$) were independently linked to severe fibrosis. No association was found between fructose intake and liver necroinflammatory activity, steatosis, and the features of NASH. CONCLUSIONS: The daily intake of industrial, not fruit fructose is a risk factor for metabolic alterations and the severity of liver fibrosis in patients with G1 CHC.